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Roll No.....

Degree: B.Tech. Semester: III - Course work
END-SEMESTER EXAMINATION, NOV.- DEC. 2024
Course Title: *Mathematics for Communication and Signal Processing*
Course Code: ECMTC301

Time: 3 Hours

Max. Marks: 50

Note: - Attempt all questions in given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any two parts of the following:		
1a	(i) Show that $G = \{(a, b) : a \neq 0, b \in \mathbb{R}\}$ is a group under the composition given by $(a, b)(c, d) = (ac, bd)$ (ii) Show that $H = \{(1, b) : b \in \mathbb{R}\}$ is a subgroup of the group G defined above.	5	CO1
1b	Define homomorphism of a group. If $f : G \rightarrow G'$ is a homomorphism then kernel of f is a normal subgroup of G	5	CO1
1c	Define Ring and Field . Prove that a finite Integral domain is a field.	5	CO1
Q 2	Attempt any two parts of the following:		
2a	Define subspace of a vector space. Is W a subspace or not? $W = \{(x, y, z) \in \mathbb{R}^3 \mid x + y = z \text{ and } x, y, z \in \mathbb{R}\}$	5	CO2
2b	Define linear independent and linear dependent vector. Are the vectors $x_1 = (1, 2, -1)$, $x_2 = (1, -2, 1)$, $x_3 = (-3, 2, -1)$, $x_4 = (-2, 0, 0)$ in $\mathbb{R}^4$ linear independent or not?	5	CO2
2c	Define linear transformation. Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by $T(x, y, z) = (x + y, x + z)$. Show that T is linear transformation.	5	CO2
Q 3	Attempt any two parts of the following:		
3a	Evaluate $\int_C \text{curl } \vec{F} \cdot \hat{n} dS$, where $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$ and S is the portion of the surface $x^2 + y^2 - 2ax + az = 0$ above the xy -plane. Also verify the stokes' theorem.	5	CO3
3b	Verify Green theorem in the xy -plane for $\int_C [(xy^2 - 2xy) dx + (x^2y + 3) dy]$ around the boundary C of the region enclosed by $y^2 = 8x$ and $x = 2$.	5	CO3
3c	Verify Divergence theorem for $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ taken over the cube bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.	5	CO3

Q 4	Attempt any two parts of the following:		
4a	Solve the following equation using by Laplace Transformation Method: (i) $x'' - 2x' + 2x = 0$, $x_0 = x_1 = 1$ (ii) $(D^2 - D - 2)x = 20 \sin 2t$, $x(0) = -1$, $x'(0) = 2$.	5	CO4
4b	Find Laplace Transformation of the following: (i) $t^2 \sin at$ (ii) $\cosh at \cos at$ (iii) $te^{-4t} \sin 3t$.	5	CO4
4c	State Convolution theorem. Show that (i) $\mathcal{L}^{-1} \left(\frac{s^2}{(s^2+a^2)(s^2+b^2)} \right) = \frac{a \sin at - b \sin bt}{a^2 - b^2}$ (ii) $\mathcal{L}^{-1} \left(\frac{s^2}{s^4 - a^4} \right) = \frac{\sinh at + \sin at}{2a}$.	5	CO4
Q 5	Attempt any two parts of the following:		
5a	Solve $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$ by separation of variables method.	5	CO5
5b	Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with boundary conditions $u(x, 0) = 3 \sin(n\pi x)$, $u(0, t) = 0$, $u(\ell, t) = 0$ where $0 < x < \ell$.	5	CO5
5c	Define the wave equation in single dimension. Solve the equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$, indicating the vibration of the string of length ℓ , fixed at both ends, given that $y(x, 0) = f(x)$, $y(0, t) = 0$, $y(\ell, t) = 0$, and $\frac{\partial y(x, 0)}{\partial t} = 0$ where $0 < x < \ell$.	5	CO5